Experience in the Same-Gender Environments and Low-Promotability Tasks*

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Abstract

We experimentally investigate how exposure to same-gender environments affects volunteering for low-promotability tasks (LPTs) in subsequent mixed-gender groups. We find that prior same-gender exposure induces an asymmetric response: men reduce volunteering by about ten percentage points in subsequent mixed-gender rounds, while women's volunteering is essentially unchanged. In addition, groups with more women coordinate more successfully. Averaged over all rounds, however, women do not volunteer more than men in mixed-gender groups, in contrast to canonical findings in the literature. In hypothetical assignment tasks, participants of both genders predominantly assign the LPT to a man, so the social-expectation channel emphasized in prior work cannot account for volunteering patterns in our context. We organize candidate interpretations into two belief-related considerations—the precision of beliefs about others' actions, captured by strategic uncertainty, and the level of empirical expectations about whether someone else will volunteer—and one preference-based consideration, gender in-group favoritism. These considerations fit different parts of the data but are not separately identified by our design.

JEL Classification: C92, J16, I21

Keywords: Low-promotability tasks, Same-gender environments, Gender differences, Strategic uncertainty, In-group favoritism

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1 Introduction

Low-promotability tasks (LPTs), such as organizing meetings, taking minutes, or handling administrative logistics, are essential for group functioning yet offer little direct reward in terms of career advancement. A substantial body of research documents that women disproportionately accept these tasks in mixed-gender workplaces, contributing to persistent gender gaps in promotions and earnings (Babcock et al., 2017a). The dominant explanation centers on social expectations: women are more frequently asked to perform LPTs and face stronger implicit pressure to accept them, whereas men are less likely to be targeted or sanctioned for declining (Babcock et al., 2017a; Bircan et al., 2025).

At the same time, evidence shows that gender gaps in volunteering largely disappear in same-gender environments (Babcock et al., 2017a), suggesting that mixed-gender contexts activate norms or beliefs that shape LPT behavior. This raises a central question that has not been examined: Do behavioral patterns developed in same-gender environments persist when individuals later move into mixed-gender settings? If early exposure to gender-homogeneous groups weakens or prevents the internalization of gendered expectations, such environments could help mitigate long-run disparities in organizational task allocation. Conversely, if individuals adjust immediately to the gender composition of a new environment, same-gender experiences may provide little lasting protection against gendered LPT norms.

We investigate this question using a sequential two-stage laboratory experiment[1] that adapts the voluntary investment game from Babcock et al. (2017a), a stylized representation of LPT allocation in teams. Participants are first assigned either to same-gender or mixed-gender groups and later transition into the opposite environment. This design allows us to isolate the effect of prior experience through between-subjects comparisons (comparing the second stage of different treatment groups), thereby ruling out simple learning effects. This enables clean identification of whether behavioral norms or expectations formed in one environment shape willingness to volunteer in the next.

Our results are as follows. Prior exposure to same-gender groups instead induces an asymmetric response: men who first interact in same-gender groups volunteer about 10 percentage points less in subsequent mixed-gender rounds than men who start in mixed-gender groups, whereas women's volunteering is essentially unchanged. This divergence produces an economically meaningful gender gap after same-gender exposure, although the corre-

[1]This experiment and the main hypotheses are pre-registered at AsPredicted.org (#236756).

sponding difference-in-differences estimate is statistically significant only at the 10 percent level in specifications that include survey controls. Averaged across all mixed-gender rounds, however, women do not volunteer at higher rates than men, in contrast to the canonical pattern documented in the literature. Our hypothetical supervisory scenarios also provide no evidence that participants reproduce the female-assignment norm emphasized in prior work: respondents of both genders predominantly assign the LPT to a man. Thus, in our South Korean sample, the behavioral patterns arise in a context where elicited assignment choices favor men rather than women, providing evidence of a boundary condition for accounts based on female-directed social expectations.

The absence of gendered assignment expectations in our context prompts us to consider three possible explanations, which fall into two broad categories: two are belief-related, and one is preference-based. The first belief-related explanation concerns differential aversion to strategic uncertainty—the uncertainty stemming from the simultaneous and unobservable choices of other group members. If beliefs about others' behavior are less precise in mixed-gender groups, and if women are more averse to this type of uncertainty, women would be more willing to volunteer in order to secure group coordination. Prior experimental research documents that women often adopt actions that stabilize collective outcomes in coordination settings. We emphasize, however, that our design does not elicit beliefs or manipulate uncertainty directly, so this interpretation remains one among several candidates rather than an established mechanism. The second belief-related explanation is revised empirical expectations about whether another group member will volunteer, while the preference-based explanation is gender in-group favoritism; both are developed in Section 5.

Among these explanations, the schooling-background patterns are most directly related to the strategic uncertainty interpretation. Women with single-sex schooling backgrounds volunteer more than women from coeducational schools, particularly in mixed-gender rounds. This association, however, is slightly larger among participants who start in mixed-gender groups, does not differ significantly by prior same-gender exposure within the experiment, and is accompanied by a similar association among men with all-boys schooling backgrounds. Because schooling background is not randomly assigned, we interpret these patterns as descriptive.

Overall, our findings challenge the view that gender differences in LPT volunteering are driven by a single universal mechanism. Social expectations, emphasized in prior literature, cannot explain our data by themselves; strategic uncertainty, revised expectations about others' behavior, and gender in-group favoritism can each rationalize parts, though not all, of the observed patterns. More broadly, the results suggest that prior experience in same-gender environments may shape how individuals navigate later mixed-gender settings, with potential implications for organizational design and policies aimed at reducing unequal task burdens.

The remainder of this paper is organized as follows. The following section provides an overview of the relevant literature on low-promotability tasks and single-gender educational environments. Section 3 describes the experimental design and hypotheses. Section 4 presents the results. Section 5 discusses the implications of our findings and proposes directions for future research. Section 6 concludes.

2 Literature Review

2.1 Low-promotability tasks

A large body of experimental research has documented persistent gender disparities in the allocation of LPTs. Babcock et al. (2017a) document that, in mixed-gender groups, women volunteer for or accept LPTs at substantially higher rates than men, whereas such differences vanish in same-gender settings. Subsequent work by Babcock et al. (2017b) extend this analysis by examining whether backlash from refusing LPTs contributes to observed gender gaps. While confirming that women are more likely to receive requests for and accept such tasks, the authors conclude that backlash neither accounts for nor amplifies these disparities.

Building on these foundational insights, more recent research has explored how institutional signals and organizational environments influence LPT behavior. In a randomized field experiment, Villas-Boas et al. (2019) provide evidence that peer-comparison information regarding service participation significantly increases volunteering rates in university departments, with effects driven primarily by women despite comparable pre-treatment behavior across genders. Complementary evidence from Banerjee and Mustafi (2025) indicate that simple social-recognition interventions can substantially raise volunteering and narrow gender gaps in willingness to perform LPTs, underscoring the malleability of gendered behavioral patterns in response to institutional design. Related studies further suggest that team gender composition shapes collaboration and cooperative behavior (Berge et al., 2016), and that leadership roles in coordination settings are evaluated and rewarded differently by gender (Grossman et al., 2019), highlighting how team context affects both participation and its perceived value.

More recent contributions have linked LPT behavior to longer-term career outcomes. Using data from knowledge-intensive workplaces, Bircan et al. (2025) show that women are systematically assigned fewer promotable tasks and more non-promotable ones, and that these task-allocation disparities explain a meaningful share of observed gender promotion gaps. Consistent patterns emerge in academic settings as well: studies such as Guarino and Borden (2017) and O'Meara et al. (2017) report that women disproportionately shoulder internal service, administrative responsibilities, and mentoring duties, closely aligning with LPT dynamics identified in experimental research. Structural and intersectional analyses, including Crimmins et al. (2023), further emphasize that the unequal distribution of non-promotable work constitutes a systemic, institution-level barrier to women's advancement.

2.2 Single-Gender Education and Competitiveness

A parallel line of research investigates the behavioral consequences of single-sex schooling, although the evidence on competitiveness remains mixed. Exploiting randomized controlled school assignment in South Korea, Lee et al. (2014) report no significant increase in competitiveness among girls attending single-sex schools relative to their co-educated peers. In contrast, evidence from Booth et al. (2014) indicate that girls assigned to all-female groups display greater willingness to take risks than those in mixed-gender environments, pointing to the importance of peer composition in shaping behavioral expression. Along similar lines, Laury et al. (2019) document that single-gender classrooms raise female students' propensity to engage in competitive tasks and to take risks. Complementary evidence further suggests that exposure to single-sex educational settings can generate persistent differences in competitive choices that remain detectable even after individuals subsequently interact in mixed-gender groups (Booth and Nolen, 2012).

Beyond short-run behavioral responses, research leveraging random assignment to single-sex versus coeducational classes indicates that single-sex schooling can influence female students' academic outcomes and self-assessments, highlighting mechanisms that may extend to later-life contexts (Eisenkopf et al., 2015). Extending this literature, Kim et al. (2024) provide evidence that exposure to single-sex schooling has effects that persist beyond adolescence, shaping women's preferences and early career-related outcomes in adulthood.

However, existing work has not examined whether these tendencies carry over when individuals transition into mixed-gender professional environments, nor whether they shape behavior specifically in the domain of LPTs.

2.3 Gender Composition, Beliefs, and Preferences

The payoff structure underlying the decision to volunteer for an LPT closely resembles the Volunteer's Dilemma (Diekmann, 1985) or the Dragon-Slayer game (Bliss and Nalebuff, 1984). In these settings, a collective good is provided only if at least one individual bears a personal cost; absent such action, all group members experience a worse outcome. Volunteering, therefore, need not reflect altruistic motives but can instead arise from concerns about coordination failure.

A related literature emphasizes the role of strategic uncertainty—uncertainty stemming from others' simultaneous choices—as a distinct determinant of behavior, separate from risk or ambiguity. Early studies of coordination games (e.g., Van Huyck et al., 1990; Cooper et al., 1992) document frequent failures to reach efficient equilibria, not because individuals are risk-averse, but because they lack confidence in others' willingness to act. More formal evidence disentangling strategic uncertainty from risk is provided by Heinemann et al. (2009), who show that individuals adopt precautionary strategies around coordination thresholds even when monetary risk is fully removed. Similar patterns emerge in bargaining environments: Greiner (2023) reports that individuals often opt for safer yet inefficient strategies in order to avoid exposure to potentially adverse choices by rivals. Complementary work distinguishing empirical beliefs about others' behavior from normative expectations further indicates that these belief components can exert independent effects on cooperation, a distinction that is especially salient in environments where others' actions are difficult to anticipate (Kölle and Quercia, 2021).

A smaller but growing body of research considers whether responses to strategic uncertainty vary by gender. In public-goods settings, evidence from Croson and Gneezy (2009) and Chaudhuri (2011) suggests that women tend to adopt more predictable contribution strategies, particularly when group behavior is uncertain. In threshold and weakest-link games, Dufwenberg and Gneezy (2005) observe that gender differences in coordination are not ubiquitous, but are more likely to arise when beliefs about others' actions are imprecise. More recent studies offer sharper evidence of context-dependent gender effects. For instance, Chang et al. (2024) show that making gender identity salient in stag-hunt games increases women's cooperative choices, consistent with identity serving as a coordination device that mitigates strategic uncertainty. Similarly, Cason et al. (2022) report that groups with a higher share of women are more likely to select prosocial equilibria, an outcome attributed to stronger beliefs in mutual coordination rather than to differences in risk preferences.

In such settings, women more frequently choose actions that stabilize collective outcomes, thereby limiting their exposure to coordination failure. Notably, this insight has not yet been applied to the context of LPTs, which similarly require unilateral and predictable action to ensure group efficiency, nor has prior work examined how transitions between homogeneous and heterogeneous social environments may generate gender-specific responses to strategic uncertainty.

A related consideration is that both beliefs and social preferences may depend on group gender composition. Beliefs about others' behavior respond to the gender of one's counterparts: Holm (2000) shows that gender operates as a focal point in coordination games, with players of both genders adopting more aggressive strategies against female counterparts, and Babcock et al. (2017a) documents that both men and women believe a woman is more likely to volunteer. Social preferences may likewise depend on group composition. Group-identity experiments show that individuals attach greater weight to in-group members' payoffs (Chen and Li, 2009), women cooperate more in all-female groups (Croson et al., 2008), and automatic gender in-group bias is markedly stronger among women than among men (Rudman and Goodwin, 2004). The evidence is not uniform—same-gender pairs can bargain less efficiently (Sutter et al., 2009), and dictator transfers between women are not always larger (Ben-Ner et al., 2004)—but it suggests that gender-homogeneous and gender-heterogeneous environments may differ not only in the predictability of others' actions but also in the weight individuals attach to others' payoffs.

2.4 Gap in the Literature and Contribution

Work on LPTs has primarily attributed women's higher volunteering in mixed-gender groups to social expectations and the interpersonal costs associated with refusing requests (Babcock et al., 2017a,b), while related evidence emphasizes that team context shapes both cooperative dynamics and perceptions of responsibility, including through group gender composition and gendered evaluations of leadership (Berge et al., 2016; Grossman et al., 2019). Separately, research on single-sex education documents persistent effects of gender-homogeneous exposure that remain observable even after individuals later transition into mixed-gender settings (Booth and Nolen, 2012; Eisenkopf et al., 2015). A distinct literature on cooperation under strategic uncertainty, in turn, highlights the central role of beliefs about others' actions and shows that normative expectations can diverge from empirical beliefs when uncertainty is high (Kölle and Quercia, 2021). Yet no study brings these mechanisms together in an LPT-like Volunteer's Dilemma environment or examines whether prior same-gender exposure shapes volunteering behavior once individuals enter mixed-gender groups.

Our study bridges these strands through a sequential design that varies group gender composition over time and directly elicits assignment expectations. We show that a gender gap in volunteering emerges following same-gender exposure even when assignment expectations are absent or reversed, and that this gap is larger among women originating from more predictable same-gender environments. These findings are consistent with a strategic uncertainty account in which volunteering serves as a stabilizing action that mitigates coordination failure, and they underscore how early social environments condition responses to heterogeneity in later interactions.

3 Experiment

To examine whether behavioral patterns shaped in same-gender settings persist in subsequent mixed-gender settings, we designed a two-stage laboratory experiment where the main task is to play the investment game used in Babcock et al. (2017a). Each stage varied the gender composition of participants' groups, allowing us to identify carry-over effects while holding the task structure constant.

3.1 Experimental Design

An experiment is designed as follows: In each session, participants completed 14 rounds of a three-person investment game. In every round, participants were randomly regrouped into triads and simultaneously[2] chose whether to invest. The payoff structure is as follows:

- If no one invests, all earn 100 tokens;
- If one invests, that individual earns 125 tokens and the others earn 200 each; and
- If multiple participants choose to invest, one participant is randomly selected to be the investor who earns 125 tokens, and the others earn 200 each.

Although framed as an investment game, the payoff structure resembles that associated with the LPTs: the participants are asked to decide whether to volunteer for a task that

[2]While some prior LPT experiments utilize a continuous-time setup with a countdown timer (e.g., Babcock et al., 2017a), we utilize a static, simultaneous-move game. A normal-form game is theoretically identical to a sequential game with imperfect information. Practically, this design choice eliminates potential confounds related to reaction times or countdown-induced urgency, allowing for a cleaner elicitation of strategic uncertainty aversion.

benefits the group but yields little personal reward, while the group earns lower payoffs unless someone volunteers on the task. As the experimental setup shares the critical features of the LPT setup, we interchangeably interpret volunteering to invest as volunteering for an LPT.

While we adapt this core LPT framework, we implemented specific methodological controls tailored to our research question. Because our primary objective is to isolate how gender composition of a group alters volunteering, participants received no feedback regarding group outcomes between rounds, and payment was based on a single randomly selected round at the end of the experiment. These design choices are crucial for preventing dynamic wealth effects, strategic hedging, and potential cross-round contamination (such as income effects or learning from the success of previous, differently composed groups). A further concern specific to our setting is that round-level feedback would have served as an indirect signal about group composition and thereby shaped participants' beliefs in ways that vary systematically across conditions.[3] Because the informational content of feedback is itself composition-dependent, providing feedback would have endogenously shifted beliefs about others' behavior across treatments, confounding the comparison our design seeks to identify. By eliminating these potential biases, our design ensures that each decision strictly and independently captures the participant's response to their immediate social environment.

Participants were randomly assigned to one of two treatment conditions. In the *SameFirst* treatment, subjects first completed 6 rounds of the investment game in same-gender groups followed by 8 rounds in mixed-gender groups. In the *MixedFirst* treatment, this order was reversed.

The most practical challenge was to make the gender composition of the group salient without raising unnecessary experimenter demand effects. Among several alternatives we could consider,[4] we decided to implicitly inform the gender composition through silhouettes

[3]For example, a male participant in a 2-women-1-man group who chose not to invest and then learns that someone else did can infer with certainty that a woman invested. The same feedback conveys substantially weaker gender-specific information in a 2-men-1-woman group.

[4]Making some characteristics of the group members salient without an unnecessary emphasis is easier said than done, as these are on a clear tradeoff relationship. Alternative designs considered in other studies and the reason why we did not adopt are as follows. Croson et al. (2008) composed the single-gender sessions by recruiting subjects from fraternities and sororities, while the both-gender sessions were recruited from a service fraternity and a business fraternity. This design would innocuously allow the subject to infer with whom they participate in the experiment. In a similar vein, Brañas-Garza et al. (2006) recruited Gitano subjects and indirectly allowed them to infer that they would play a game with another Gitano from their own neighborhood. Since we want to inform subjects a specific gender composition per round, indirect information through their own inference was not viable. Solnick (2001) uses real first names to examine gender differences in the ultimatum game, and Kim and Riegel (2025) ask the participants to choose a gender-identifiable nickname

(see Figure 1). The participant's silhouette image at the center is always matched with the participant's gender. Two members' silhouettes in the group are shown on the left and the right side so that the information about gender composition can be immediately delivered. Although the screenshot in Figure 1 shows a male silhouette on the left, gender compositions and positions are randomized.

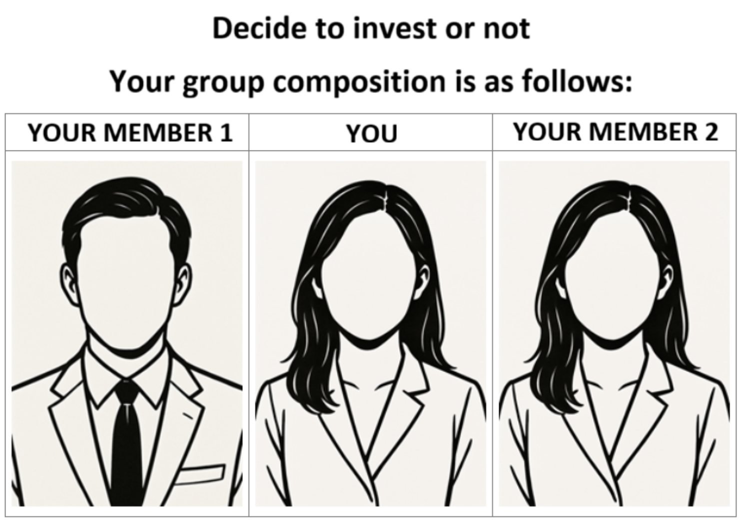

Figure 1: Screenshot of one decision round

In the *SameFirst* treatment, subjects were informed at the beginning of round 1 that they would play the investment game for the next 6 decision rounds with randomly reshuffled group of the same gender, and then at the beginning of round 7, they were again informed that they would play the game for the next 8 rounds with at least one member of a different gender.[5] In the *MixedFirst* treatment, the reversed information was given at the beginning of round 1 and round 9.

It is important to note that our primary interest lies in the transition from same-gender

for each decision round, so that the gender information of other participants was innocuously revealed. We were concerned that gender-neutral real names might be uninformative, and that both real and pseudo names could introduce uncontrolled behavioral confounds by signaling unintended characteristics like prosociality or assertiveness (Grundmann et al., 2025).

[5]We acknowledge that explicitly announcing the group composition makes gender salient, which might raise concerns regarding experimenter demand effects. However, this design choice is necessary to prevent confounding beliefs regarding the matching process. If the gender composition were left unexplained, a participant in the *SameFirst* treatment might incorrectly infer that the initial string of same-gender partners was a statistical coincidence or that the silhouette images were uninformative at all. Consequently, the sudden appearance of a mixed-gender group in round 7 could be interpreted as a software error or an outlier rather than a change in the environment. The explicit instruction ensures that behavioral changes are driven by the new social context rather than by surprise or confusion about the matching probabilities.

to mixed-gender environments (the *SameFirst* treatment). The *MixedFirst* treatment serves as a control. By observing behavior in mixed-gender groups prior to any same-gender experience, we establish a baseline that allows us to isolate the effect of prior same-gender exposure while controlling for general learning, order, and fatigue effects across the 14 rounds. It is not intended as a theoretical test of reverse transitional effects.

After the decision rounds, participants completed a post-experimental survey measuring perceived pressure to invest, expectations about same-gender group members, concerns about others' judgments, and attitudes relating to leadership, conformity, and social expectations. Crucially, to test for the presence of gendered LPT assignment norms, the survey included hypothetical assignment tasks. Participants were asked to imagine acting as an uninvolved supervisor who must instruct one group member to make the investment. To maintain consistency with the main task, they were presented with the same silhouette interface, displaying two different mixed-gender triads: one consisting of two males and one female, and another consisting of one male and two females. Participants then explicitly selected which silhouette should perform the task.[6]

3.2 Hypotheses

Drawing on the motivations and prior empirical findings reviewed above, we formulate four testable hypotheses that reflect distinct mechanisms through which gender composition may influence willingness to volunteer for LPTs. These hypotheses address (i) baseline gender differences in mixed-gender environments, (ii) the persistence of behavioral patterns formed in same-gender settings, (iii) long-term effects of single-sex schooling, and (iv) the role of gendered expectations in LPT assignment.

Hypothesis 1. *Women volunteer for LPTs at higher rates than men in mixed-gender environments.*

Hypothesis 1 establishes a behavioral benchmark within our sequential experimental framework. Prior studies document higher female volunteering in mixed-gender groups in static environments where gender composition is fixed throughout the interaction. In contrast, our design introduces dynamic changes in group composition, making it unclear ex ante whether the canonical mixed-gender gender gap should emerge.

Sequential exposure may alter beliefs, dampen social expectations, or induce anticipatory behavior that attenuates contemporaneous gender effects. As a result, the presence or

[6] The full survey instrument is provided in Appendix B.

absence of a gender gap in mixed-gender rounds is itself informative. Hypothesis 1 therefore does not merely replicate earlier findings; it tests whether gender composition remains behaviorally salient when embedded in a dynamic environment that allows for experience-based belief formation.

Establishing this benchmark is also essential for interpreting subsequent hypotheses. In particular, Hypothesis 2 examines whether behavior in mixed-gender environments depends on prior exposure, and Hypothesis 3 investigates long-run effects of educational environments. Both require clarity about baseline behavior in mixed-gender groups within the sequential setting.

Hypothesis 2. *Women who initially experience in same-gender environments are less likely to volunteer for LPTs when later placed in mixed-gender groups, relative to those who first experience mixed-gender groups.*

Hypothesis 2 tests whether behavioral norms and expectations formed in same-gender settings persist when individuals transition into mixed-gender environments. Participants who begin in the *MixedFirst* treatment enter the experiment under conditions where gender heterogeneity is salient from the outset, potentially activating gendered expectations or other mechanisms associated with mixed-gender interactions. By contrast, participants in the *SameFirst* treatment initially interact under an egalitarian norm in which LPT responsibilities are not associated with gender. If early exposure shapes subsequent beliefs or sense of responsibility, then *SameFirst* women should exhibit lower volunteering rates than *MixedFirst* women after both groups are placed in the same mixed-gender environment. A null or opposite effect would suggest that behavior is more strongly influenced by the immediate social composition of the group than by prior exposure.

Hypothesis 3. *Women with single-sex schooling backgrounds are less likely to volunteer for LPTs than women from coeducational schools, particularly within mixed-gender environments.*

Hypothesis 3 extends the analysis from short-term experimental exposure to long-term educational environments. Prior work shows that single-sex schooling can influence competitiveness, risk-taking, and labor-market outcomes (Booth et al., 2014; Laury et al., 2019; Kim et al., 2024). Whether such environments shape LPT-related behavior remains an open empirical question. One possibility is that single-sex schooling reduces women's exposure to gendered expectations regarding LPTs, lowering volunteering rates. We set our hypothesis based on this reasoning. Another possibility is that single-sex schooling strengthens confidence and initiative-taking in group settings, leading women with such backgrounds to volunteer more, especially when placed in mixed-gender groups where peer predictability is lower.

Hypothesis 4. *When subjects, acting as supervisors, are asked to assign an LPT to one member in a mixed-gender group, they would choose the female member more frequently.*

Hypothesis 4 tests the prevailing social expectation mechanism documented in the literature, which suggests that the disproportionate volunteering by women may be driven by an internalized social norm that women are expected to assume these duties. If this norm holds in our context, participants acting as an uninvolved third-party (supervisor) should explicitly act on this expectation by assigning the LPT to a female group member. Critically, a rejection of this hypothesis, which would occur if participants choose men more often than its proportion, provides evidence of a cultural heterogeneity in LPT assignment norms. This result is crucial because it helps to decouple the observed volunteering behaviors (Hypotheses 1 and 2) from the social expectation mechanism, suggesting that women's higher volunteering rate might instead be driven by other factors, such as strategic uncertainty, risk preferences, or greater sensitivity to the gender composition of the group.

A few remarks are in order. Hypotheses 1 and 4 test the basic mechanisms and cross-cultural validity of the LPT literature concerning observed behavior and explicit social norms, respectively. Hypothesis 2 tests whether short-term experimental exposure to same-gender settings influences later LPT behavior in mixed-gender environments. Finally, Hypothesis 3 extends this inquiry by exploring whether long-term educational experiences shape LPT behavior beyond short-term experimental exposure. Together, these hypotheses allow us to evaluate the short-run, persistent, and long-term influences of same-gender contexts on LPT volunteering and the cultural validity of the underlying social norms.

3.3 Experimental Procedure

Participants were recruited from Korean universities and completed the experiment remotely via Zoom using the LIONESS online experimental platform. At registration, participants provided demographic information, including gender and schooling background, enabling gender-balanced session assignments. Because each treatment required precise gender ratios, unmatched participants were turned away when necessary. We ended up having 78 subjects (39 males and 39 females) for the *SameFirst* treatment, and 72 subjects (36 males and 36 females) for the *MixedFirst* treatment.

Over the 14 rounds, no feedback was provided regarding group members' decisions or payoffs, ensuring that learning occurred only through belief formation rather than outcome observation. The participants receive cash-equivalent mobile payment credits corresponding to the payoff from one randomly selected round.

4 Results

This section presents the experimental findings by examining investment behavior at the individual and group levels. Across all treatments, groups successfully coordinated an investment 73.81 percent of the time, slightly lower than in Babcock et al. (2017a) but substantially above the mixed-strategy equilibrium prediction. Balance tests indicate that observable characteristics are similar across treatments, except that *SameFirst* participants report more acquaintances in the session; see Appendix Table A.1.[7]

Table 1: Investment rates and within-person changes by treatment, gender, and phase

	N	Phase 1	Phase 2	Δ (pp)	p
Panel A. SameFirst (same → mixed-gender)					
Men	39	35.9	32.1	−3.8	0.285
Women	39	38.9	40.4	+1.5	0.510
Panel B. MixedFirst (mixed → same-gender)					
Men	36	38.9	38.4	−0.5	0.935
Women	36	37.8	50.0	+12.2	0.052

Notes: Investment rates in percent. In SameFirst, Phase 1 covers rounds 1–6 (same-gender) and Phase 2 rounds 7–14 (mixed-gender); in MixedFirst, Phase 1 covers rounds 1–8 (mixed-gender) and Phase 2 rounds 9–14 (same-gender). Δ is the average within-person change in each subject's investment proportion from Phase 1 to Phase 2. p-values are from paired t-tests with one observation per subject per phase, which account for within-subject correlation across rounds by construction. Round-level regressions of the investment decision on a Phase-2 indicator with standard errors clustered by subject yield identical point estimates and nearly identical p-values. We cluster standard errors at the subject level throughout, as in Babcock et al. (2017a).

Table 1 reports raw investment rates by treatment, gender, and phase, together with within-person changes across phases. Three observations stand out. First, phase-1 investment rates are similar across the four groups, ranging from 35.9 to 38.9 percent. Second, within-person changes upon the environmental transition are small and statistically insignificant for three of the four groups: *SameFirst* men (−3.8 percentage points), *SameFirst* women (+1.5), and *MixedFirst* men (−0.5). Third, the one sizable change is that *MixedFirst* women increase their investment by 12.2 percentage points upon moving from mixed-gender to same-gender groups ($p = 0.052$). We return to this pattern in Section 5, as it disciplines the set of mechanisms that can account for our results.

[7] Controlling for the number of acquaintances does not qualitatively change the results.

The next two figures show the overall tendency of investment by gender. Note that these figures are for illustrating the overall uncontrolled patterns, so the inference from them should be limited. Figure 2 shows the investment rate by gender in the *SameFirst* treatment. Females seem to invest a bit more than males in the later (mixed-gender) rounds, but the difference is not statistically significant ($p = 0.284$).[8] This null result pertains even after controlling for rounds and other control variables: see Appendix Table A.3.

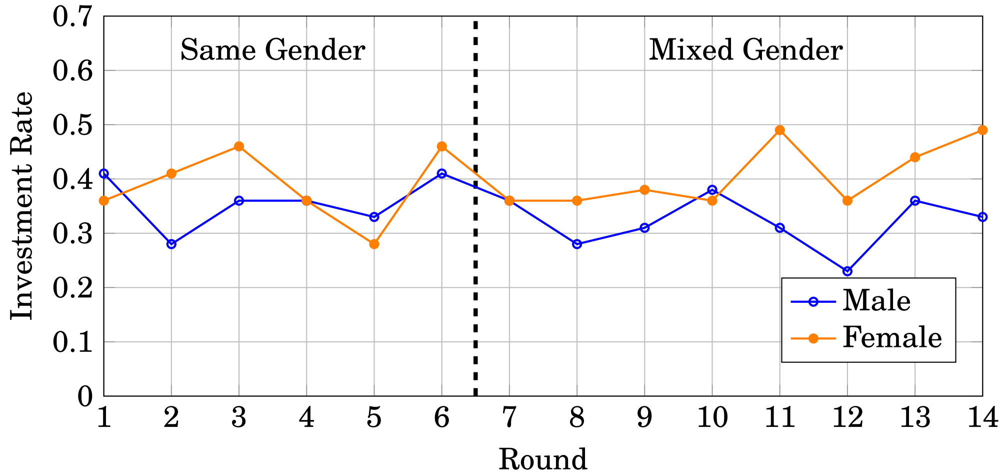

Figure 2: Investment Rate over Round by Gender, *SameFirst* Treatment

Figure 3 shows the investment rate by gender in the *MixedFirst* treatment. Females seem to invest more than males in the later (same-gender) rounds, but this difference is again not statistically significant ($p = 0.215$) even after controlling for other observable variables: see Appendix Table A.4. Across the two treatments, we find no evidence that women volunteer for LPTs at higher rates than men in mixed-gender environments and therefore no support for Hypothesis 1. The estimate is imprecise, however, and its confidence interval does not rule out economically meaningful gender differences.

Result 1. *Overall, women do not volunteer for LPTs at higher rates than men in mixed-gender environments.*

This result suggests that the interaction of gender and prior exposure is worth examining. Table 2 reports, for each subsample, average adjusted probabilities of investing for the

[8] Unless otherwise stated, we report results from Probit regressions with standard errors clustered at the individual level. Main tables report average adjusted probabilities for each treatment-by-gender cell, with covariates held at observed values, and contrasts between cells computed as linear combinations of the cell probabilities with delta-method standard errors. For the interaction of two binary variables, the reported difference-in-differences is the cross-difference of predicted probabilities (Ai and Norton, 2003). Results from linear probability model regressions are qualitatively similar and available upon request.

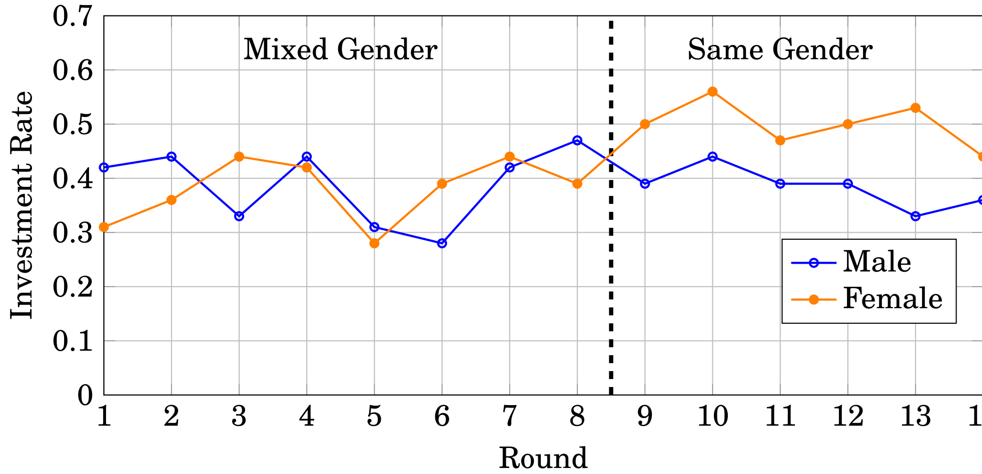

Figure 3: Investment Rate over Round by Gender, *MixedFirst* Treatment

four treatment-by-gender cells (Panel A) and the contrasts between cells that correspond to the comparisons of interest (Panel B). Our main focus is on the mixed-gender subsample, columns (5) and (6). Three findings emerge. First, prior same-gender exposure is associated with lower investment among men: *SameFirst* men invest 10.1 percentage points less than *MixedFirst* men in mixed-gender rounds ($p = 0.240$; 14.0 points with survey controls, $p = 0.057$). Second, the corresponding effect for women is essentially zero (-0.09 points, $p = 0.905$). Third, as a consequence of this asymmetry, the gender gap in mixed-gender rounds is larger in *SameFirst* than in *MixedFirst*: the difference-in-differences is 9.2 percentage points ($p = 0.388$; 17.7 points with survey controls, $p = 0.063$), and the gender gap within *SameFirst* is 8.2 points ($p = 0.282$; 11.5 points with controls, $p = 0.098$). Because the survey controls are measured after the experiment and may themselves respond to treatment, we treat the specifications without controls as primary and those with controls as robustness checks.[9] Two conclusions follow. Prior same-gender exposure does not reduce women's subsequent volunteering, so Hypothesis 2 finds no support; and the reduction in overall volunteering after same-gender exposure is driven entirely by men.[10]

[9]The survey controls are five-point Likert items from the post-experiment questionnaire (e.g., "I feel pressure to invest," "I expect someone else with the same gender of mine would invest"); full item wordings are reproduced in Appendix B.

[10]The effect of the number of played rounds (Round) is consistently insignificant across our main specifications, indicating the absence of general learning or fatigue effects over time. Table A.2 provides an additional robustness check focusing strictly on Rounds 7 and 8, which holds the level of game experience constant across treatments. The asymmetry is consistent: upon entering the mixed-gender environment, *SameFirst* men invest 12.4 percentage points less than *MixedFirst* men ($p = 0.196$; 17.3 points with survey controls, $p = 0.028$), while the corresponding difference for women is essentially zero.

Table 2: Investment by treatment and gender: cell probabilities and contrasts

	Pooled Data (All Rounds)		**Same Environment Only**		**Mixed Environment Only**	
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Panel A. Average adjusted probabilities						
SameFirst × Female	0.397	0.411	0.373	0.381	0.387	0.406
SameFirst × Male	0.337	0.324	0.344	0.335	0.306	0.291
MixedFirst × Female	0.430	0.403	0.518	0.488	0.396	0.370
MixedFirst × Male	0.387	0.414	0.402	0.429	0.407	0.432
Panel B. Contrasts						
SameFirst effect, women ($\mu_{11} - \mu_{01}$)	−0.0331	+0.0077	−0.1447	−0.1068	−0.0088	+0.0364
	(0.621)	(0.904)	(0.150)	(0.289)	(0.905)	(0.606)
SameFirst effect, men ($\mu_{10} - \mu_{00}$)	−0.0499	−0.0901	−0.0579	−0.0944	−0.1012	-0.1404^{*}
	(0.526)	(0.167)	(0.598)	(0.348)	(0.240)	(0.057)
Gender gap, SameFirst ($\mu_{11} - \mu_{10}$)	+0.0603	+0.0870	+0.0295	+0.0459	+0.0818	$+0.1147^{*}$
	(0.412)	(0.191)	(0.692)	(0.504)	(0.282)	(0.098)
Gender gap, MixedFirst ($\mu_{01} - \mu_{00}$)	+0.0435	−0.0108	+0.1163	+0.0583	−0.0106	−0.0621
	(0.547)	(0.867)	(0.214)	(0.491)	(0.888)	(0.368)
Difference-in-differences	+0.0168	+0.0978	−0.0869	−0.0124	+0.0924	$+0.1768^{*}$
	(0.871)	(0.277)	(0.467)	(0.907)	(0.388)	(0.063)
Survey controls	No	Yes	No	Yes	No	Yes
Observations	2100	2100	900	900	1200	1200

Notes: Panel A reports average adjusted predictions from Probit regressions of the investment decision on treatment, gender, their interaction, and round, with covariates held at observed values and treatment and gender set to the indicated cell. Panel B reports linear combinations of Panel A cells; μ_{tf} denotes the cell probability for treatment t (1 = *SameFirst*) and gender f (1 = female). The difference-in-differences row is the Ai and Norton (2003) cross-difference. p-values based on delta-method standard errors clustered at the individual level are reported in parentheses. Even-numbered columns add post-experimental survey controls; because these are measured after treatment, they serve as robustness checks. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Result 2. *Prior same-gender exposure induces an asymmetric response: men reduce volunteering in subsequent mixed-gender groups while women do not. The resulting gender gap after same-gender exposure is positive but not statistically significant at conventional levels.*

The asymmetry in Result 2—women maintain their volunteering while men withdraw—leads us to a next question: is this pattern more pronounced for females who have single-gender schooling experience than for females with coeducation backgrounds? Figure 4 shows the investment rate by single-gender schooling experience over rounds, female only. The left panel depicts observations from the *SameFirst* treatment, while the right one from the *MixedFirst* treatment. Except for the same-gender environment (first 6 rounds) in the *Same-First* treatment, females with single-gender school experience tend to invest more than females from coeducational school backgrounds.

The observation from Figure 4 is examined in Table 3, which reports average adjusted probabilities and contrasts for female participants by treatment and schooling background. Girl School is a dummy variable indicating that the subject graduated from either a single-gender middle or high school. Women with girl-school backgrounds invest more than women from coeducational schools in mixed-gender rounds: the contrast is 8.1 percentage points among *SameFirst* women ($p = 0.482$) and 11.1 points among *MixedFirst* women ($p = 0.169$;

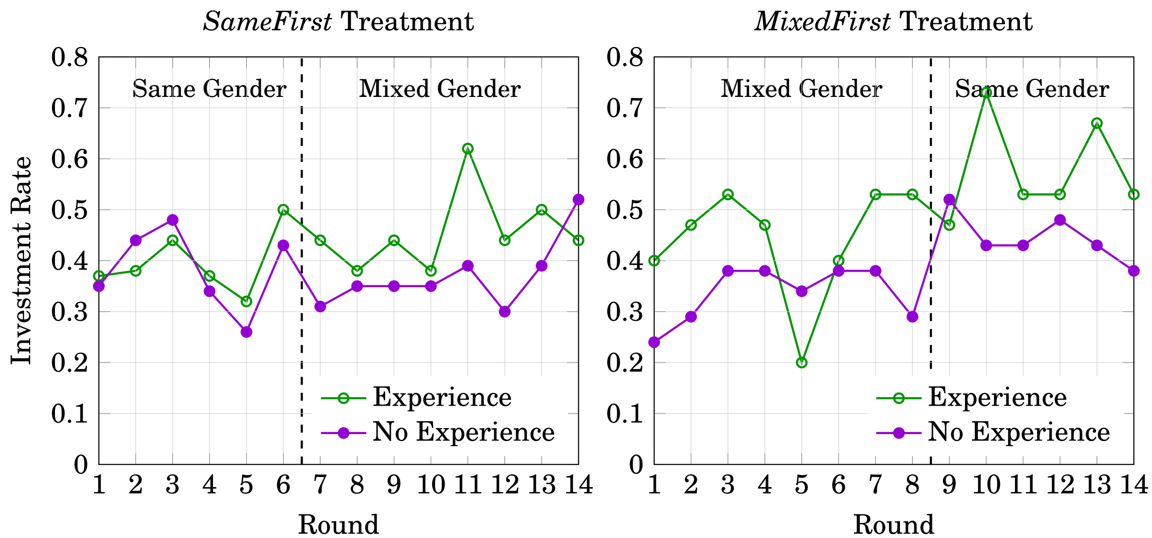

Figure 4: The effect of single-sex schooling experience, female only

16.2 points with survey controls, $p = 0.041$).[11]

Two qualifications are in order. First, the association is concentrated among *MixedFirst* participants, and the difference-in-differences between treatments is close to zero; the data therefore do not indicate that experimental same-gender exposure amplifies the schooling association. Second, the contrasts in the specifications without survey controls do not reach conventional significance levels, so the evidence is suggestive. Moreover, Appendix Table A.5 shows a qualitatively similar positive association between all-boys schooling and volunteering among men, largely within same-gender environments. The presence of a similar association for both genders cautions against a gender-specific interpretation. Because schooling background is not randomly assigned, we interpret these patterns as descriptive.[12]

Result 3. *Women with single-sex schooling backgrounds volunteer for LPTs more than women*

[11]The association is also not monotone in the number of single-sex schools attended: women who attended either a single-sex middle school or a single-sex high school, but not both, invest most in mixed-gender rounds (co-ed: 35.2 percent, $n = 44$; one single-sex school: 47.6 percent, $n = 21$; both: 38.8 percent, $n = 10$). However, only ten women attended both, so this intensity comparison is severely underpowered and we do not draw conclusions from it.

[12]Although single-gender school attendance in Seoul is not fully exogenous, it is not simply the outcome of unrestricted parental choice either. Earlier studies often treated assignment under Korea's equalization system as approximately random within districts, whereas later work noted that commuting constraints and residential sorting may weaken that assumption. More recent evidence using students' revealed school preferences from the Seoul School Choice Program suggests that, after accounting for selection into school type, single-gender schooling still has positive academic effects for girls, though not for boys (Lee and Park, 2026). We therefore interpret single-gender school experience cautiously, not as a clean causal measure, but as a substantively meaningful proxy for prior exposure to a gender-segregated school environment.

Table 3: Investment of female participants, by single-sex schooling background

	Pooled Data (All Rounds)		**Same Environments Only**		**Mixed Environments Only**	
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Panel A. Average adjusted probabilities						
SameFirst × Girl School	0.429	0.468	0.380	0.422	0.417	0.458
SameFirst × Co-ed	0.376	0.371	0.369	0.355	0.336	0.339
MixedFirst × Girl School	0.500	0.498	0.595	0.577	0.482	0.492
MixedFirst × Co-ed	0.381	0.360	0.462	0.462	0.371	0.330
Panel B. Contrasts						
SameFirst effect, Girl-School women	−0.0714	−0.0299	−0.2149	−0.1552	−0.0645	−0.0343
	(0.526)	(0.781)	(0.177)	(0.327)	(0.585)	(0.762)
SameFirst effect, Co-ed women	−0.0048	+0.0110	−0.0936	−0.1070	−0.0353	+0.0084
	(0.953)	(0.884)	(0.485)	(0.440)	(0.700)	(0.925)
Girl School effect, SameFirst	+0.0525	+0.0967	+0.0117	+0.0666	+0.0813	+0.1193
	(0.639)	(0.336)	(0.913)	(0.506)	(0.482)	(0.254)
Girl School effect, MixedFirst	+0.1192	+0.1376*	+0.1329	+0.1147	+0.1105	+0.1619**
	(0.145)	(0.074)	(0.272)	(0.315)	(0.169)	(0.041)
Difference-in-differences	−0.0666	−0.0409	−0.1213	−0.0482	−0.0292	−0.0426
	(0.631)	(0.749)	(0.453)	(0.754)	(0.835)	(0.749)
Survey controls	No	Yes	No	Yes	No	Yes
Observations	1,050	1,050	450	450	600	600

Notes: Female participants only. Girl School indicates graduation from an all-girls middle or high school; schooling background is not randomly assigned and the estimates are descriptive associations. Specifications and computation of cell probabilities, contrasts, and p-values as in Table 2. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

from coeducational schools, particularly within mixed-gender environments. This association is concentrated among MixedFirst participants and does not differ significantly by prior same-gender exposure.

Taking the first three results together, we report that (1) on average, females do not volunteer for LPTs more than males; (2) prior same-gender exposure lowers male volunteering in mixed-gender environments while leaving female volunteering unchanged, so that a statistically weak gender gap emerges after same-gender exposure; and (3) girl-school backgrounds are associated with higher female volunteering, though not differentially so after same-gender exposure. A natural question is whether these patterns reflect internalized social norms about who should perform LPTs.

If this norm holds in our context, participants should explicitly act on this expectation by assigning the LPT to a female group member when asked to choose one volunteer as an uninvolved third-party (supervisor). Specifically, when three members consist of 2 females and 1 male (or 1 female and 2 males), a female member would have been chosen more than 2/3 (or 1/3), the chances of a random selection. We find the opposite. Both males and females request a hypothetical male subject to invest.

Figure 5 shows how frequently participants, acting as supervisors of a three-person group, assign the investment task to a male group member. When choosing among two men and one woman, 77.3% of male respondents and 76.0% of female respondents assign

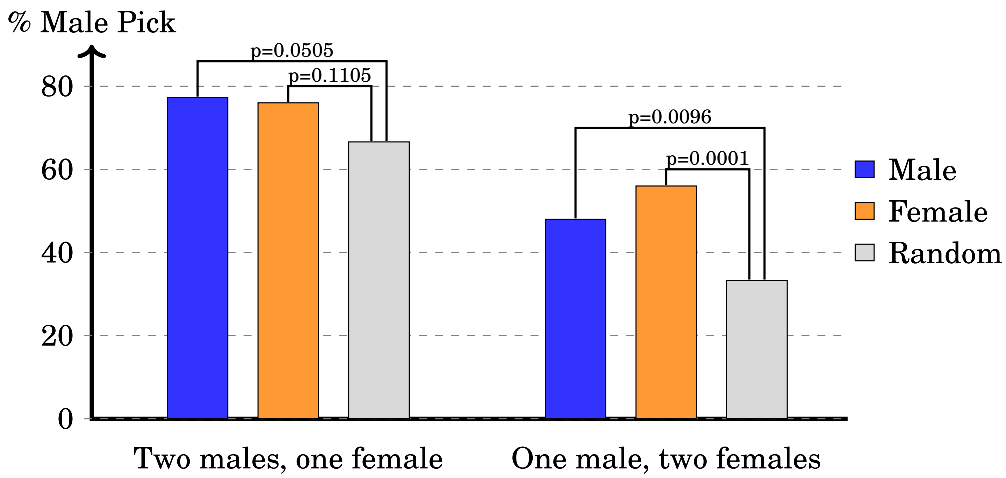

Figure 5: Assignment to Volunteer for LPTs
Notes: p-values of binomial tests are shown above bars.

the task to a man, compared with 66.7% under random assignment. When choosing among one man and two women, 48.0% of male respondents and 56.0% of female respondents assign the task to the man, compared with 33.3% under random assignment. Thus, all four comparisons run counter to Hypothesis 4. Three differ significantly from the random benchmark at the 10 percent level or better; the remaining comparison, for female respondents in the two-male-one-female scenario, points in the same direction but is not statistically significant ($p = 0.110$).[13]

Result 4. *When subjects, acting as supervisors, are asked to assign an LPT to one member in a mixed-gender group, they choose the male member more frequently.*

Two noteworthy implications are derived from Result 4. First, it provides evidence of a cultural heterogeneity in LPT assignment norms. Second, this result is crucial because it helps to decouple the observed volunteering behaviors of women from the social expectation mechanism, suggesting that women's higher volunteering rate might instead be driven by other factors, such as strategic uncertainty, expectations about others' behavior, or in-group favoritism toward same-gender peers.

[13]Male silhouettes always occupied the leftmost position in these scenarios, so a mechanical position bias could in principle inflate male choices. Among respondents who chose a male in the two-male scenario, the split between the left and the middle male silhouettes is 49.6 versus 50.4 percent ($p = 1.00$), which speaks against a simple left-position bias. Among respondents who chose a female in the two-female scenario, however, the middle silhouette is chosen more often than the right one (73.6 percent, $p < 0.001$), so position neutrality does not hold in general. The elicitation is also hypothetical and unincentivized; we therefore interpret Result 4 as evidence on stated assignment norms rather than on incentivized beliefs.

5 Discussions

This study examines whether experiences in same-gender environments shape subsequent willingness to volunteer for LPTs in mixed-gender groups. The results yield three main insights. First, on average, there is no gender gap in volunteering in mixed-gender groups. Second, prior exposure to same-gender environments reduces men's subsequent volunteering while leaving women's essentially unchanged, so that a gender gap emerges after same-gender exposure, though it is not statistically significant at conventional levels. Third, these patterns arise in a context where we find no evidence of the female-assignment expectation emphasized in prior work: participants of both genders predominantly assign the hypothetical LPT to men. Collectively, the findings indicate an asymmetric behavioral response to prior environments that cannot be explained solely by the persistence of a female-directed assignment norm.

5.1 Behavioral Persistence without Norm Persistence

The reduction in volunteering following same-gender exposure suggests that early social environments shape subsequent behavior in a path-dependent manner. This reduction, however, is driven by men: men who initially interact in same-gender groups appear less inclined to unilaterally ensure group success once they transition into mixed-gender settings, whereas women's volunteering is essentially unchanged. This pattern is consistent with prior exposure shifting men's default sense of responsibility or necessity in the group, rather than with the activation of gender-specific expectations directed at women.

Crucially, our assignment-task results show that the canonical social-expectation mechanism emphasized in the LPT literature does not operate in our setting. When asked to assign the LPT as uninvolved supervisors, both male and female participants predominantly select a male group member. This finding directly contradicts the notion that women volunteer more because they expect to be asked, sanctioned, or judged more harshly for refusing. As such, while behavior formed in same-gender environments persists, it cannot be attributed to the persistence of gendered norms regarding who should perform low-promotability work. In other words, their behavior was not driven by compliance with gender norms. The next subsection discusses three candidate drivers consistent with this pattern: aversion to strategic uncertainty, revised expectations about others' behavior, and gender in-group favoritism.

5.2 Belief- and Preference-Based Interpretations

One possible interpretation of the observed patterns concerns differential responses to strategic uncertainty. Because the design withholds round-level feedback, participants cannot update their beliefs by observing others' choices across rounds. Therefore, the announced group composition is the principal new information available for forming expectations about others' behavior. If mixed-gender groups make those expectations less precise, particularly for individuals accustomed to homogeneous environments, participants who are more averse to such uncertainty may be more willing to volunteer in order to prevent coordination failure.

Under this interpretation, choosing to volunteer functions as a form of insurance against coordination failure: it guarantees that the group avoids the 100-token outcome, while exposing the participant to the possibility of being selected as the investor and receiving 125 tokens rather than 200. Prior experimental research similarly suggests that groups containing women can be more effective at coordinating on secure outcomes (Eckel and Füllbrunn, 2015; Cason et al., 2022). The hypothetical assignment choices in Result 4, however, show that participants predominantly assign the investment task to a man. The volunteering patterns therefore do not simply reproduce a stated tendency to assign the task to women.

Two patterns limit how far a pure strategic uncertainty interpretation can be taken. First, the largest within-person change is the increase in volunteering among *MixedFirst* women upon entering same-gender groups (Table 1), whereas a simple composition-based uncertainty account would predict less volunteering in the presumably more predictable environment. Second, women with single-sex schooling backgrounds do not volunteer more in mixed-gender than in same-gender rounds within subject. Existing evidence also does not establish that mixed-gender interactions are necessarily less predictable: gender can serve as a focal point that coordinates expectations (Holm, 2000), and same-gender interactions are not always associated with more predictable or efficient outcomes (Sutter et al., 2009).

The patterns can instead be organized around three considerations within two broad categories. The two belief-related considerations concern the precision of empirical expectations, captured by strategic uncertainty, and their level, meaning the perceived probability that another group member will volunteer. The preference-based consideration is gender in-group favoritism.

For women, gender in-group favoritism may be relevant because such favoritism appears to be stronger among women than among men (Rudman and Goodwin, 2004; Croson et al., 2008). In a Volunteer's Dilemma, in-group favoritism can operate through payoff weights. If an individual places greater weight on in-group members' payoffs (Chen and Li, 2009), preventing a coordination failure that harms those members can justify a greater personal cost. A transition from a mixed-gender group to an all-female group both increases the number of female in-group beneficiaries and, through contrast with the preceding rounds, may make gender identity more salient (Chang et al., 2024). This interpretation is consistent with the increase in volunteering among *MixedFirst* women after the transition. By comparison, *SameFirst* women begin in all-female groups without such a preceding contrast and show no comparable elevation. The same welfare-weight interpretation is also consistent with Result 4: a hypothetical supervisor who places greater weight on women's payoffs may prefer to assign the investment cost to a male group member. This assignment task, however, elicits stated choices rather than incentivized beliefs or payoff weights.

For men, the in-group-favoritism consideration may be comparatively weak. Automatic gender in-group bias is substantially weaker among men, whose attitudes toward the female out-group may even be favorable (Rudman and Goodwin, 2004). Men's responses may instead reflect the level of their empirical expectations about others' behavior. These expectations concern who will act, as distinguished from normative expectations about who should act (Kölle and Quercia, 2021); Result 4 is closer to the latter because it elicits stated assignment choices. In a Volunteer's Dilemma, a higher perceived probability that someone else will volunteer reduces one's own incentive to do so. If men entering mixed-gender groups come to expect that another member will volunteer—for example, a female member, as in the belief data reported by Babcock et al. (2017a)—their own volunteering may decline, consistent with the response of *SameFirst* men. For *MixedFirst* men moving to all-male groups, the framework provides no clear prediction, and their behavior is essentially unchanged.

Table 4 summarizes how the three considerations could relate to the four transition cells. For women moving from same-gender to mixed-gender groups, greater strategic uncertainty may increase volunteering, while the loss of in-group beneficiaries may reduce it; these opposing forces are consistent with the small observed change. For women moving from mixed-gender to all-female groups, reduced uncertainty may lower volunteering in isolation, while stronger and more salient in-group considerations may raise it. Under this interpretation, the latter force would have to dominate. For men, the same-to-mixed transition may increase the perceived probability that someone else will volunteer, while the mixed-to-same transition implies no clear belief change.

This framework is an ex post organizing device rather than a test of the underlying mechanisms. None of the three considerations is directly measured, and the observed outcomes themselves are used to infer the relative directions and strengths of the proposed forces.

Table 4: Possible implications of the candidate considerations across transition cells

Transition	Gender	Belief-related consideration	Preference-based consideration	Observed
Same → Mixed	Men	expect others to act (↓)	not modeled	−3.8pp
Same → Mixed	Women	uncertainty rises (↑)	fewer in-group beneficiaries (↓)	+1.5pp
Mixed → Same	Men	no clear change	not modeled	−0.5pp
Mixed → Same	Women	uncertainty falls; free-ride (↓)	maximal and salient (↑↑)	+12.2pp

Notes: Arrows denote the direction in which each consideration may affect individual volunteering upon the transition. Observed values are the within-person changes reported in Table 1; only the change for *MixedFirst* women is statistically distinguishable from zero at the 10 percent level ($p = 0.052$). For men, "not modeled" indicates that the preference-based consideration is treated as comparatively weak and omitted from this illustrative accounting; it does not imply that male in-group favoritism is empirically zero. None of the three considerations is directly measured, and the framework is presented as an organizing device rather than as a test.

5.3 Behavioral Margins and Coordination Outcomes

We next examine whether the individual-level patterns are reflected in the distribution and timing of volunteering and in group-level coordination outcomes. Because no feedback is provided, group members cannot observe or respond to others' volunteering. Group success therefore depends on individual behavior only through the requirement that at least one member volunteers, and a group containing a near-certain volunteer succeeds regardless of the choices of the remaining members.

The clearest distributional change occurs among *MixedFirst* women. The share volunteering in at least five of six rounds rises from 6 percent in the mixed-gender phase to 36 percent in the same-gender phase, with eleven entries into the near-always state and no exits (exact McNemar $p = 0.001$). The corresponding shares are essentially unchanged among *SameFirst* participants of both genders (Appendix Table A.6, Panel A).

This shift is not confined to women with a high initial propensity to volunteer. When women are classified prospectively according to their mixed-phase behavior, both halves of the *MixedFirst* distribution increase upon entering all-female groups: volunteering rises from 63 to 71 percent among those above the median and from 22 to 36 percent among those below the median ($p = 0.093$). The corresponding changes among *SameFirst* women are only (+1.4) and (+1.6) percentage points, making simple mean reversion an unlikely explanation (Panel B).

Among the thirteen *MixedFirst* women who reach the near-always state, mixed-phase volunteering ranges from zero to eight of eight rounds, and only two were already near-always volunteers before the transition. The behavioral break is immediate: 85 percent volunteer in round 9, the first all-female round. The remaining *MixedFirst* women show no upward trend across subsequent rounds (-2.6 percentage points per round, $p = 0.26$). In the other three treatment-by-gender cells, near-always volunteers are already highly likely to volunteer at the beginning of phase 1, so no comparable low-to-high transition appears.[14]

Post-experiment survey responses provide limited additional evidence. Among *MixedFirst* women, agreement with the statement that someone of one's own gender would volunteer averages 2.38 among near-always volunteers, 3.00 among intermediate volunteers, and 3.43 among never-volunteers (Appendix Table A.6, Panel C; Spearman trend ($p = 0.11$)). The ordering is directionally consistent with greater volunteering among women who report weaker expectations that same-gender peers will act. The gradient is not present among *SameFirst* women. Because these responses were collected after the decision rounds, they may partly rationalize rather than explain behavior, and the relevant cells are small. We therefore treat this decomposition as exploratory.

Group-level coordination outcomes provide complementary evidence. We reconstruct one observation for each group-round from session, round, group composition, and outcome and estimate coordination success as a function of group gender composition. Success equals one when at least one group member volunteers. Table 5 reports average adjusted success probabilities and the corresponding contrasts.

Two findings emerge. In mixed-gender rounds, groups with two women succeed more often than groups with one woman: the difference is 15.4 percentage points among *SameFirst* participants ($p = 0.001$) and 11.4 points among *MixedFirst* participants ($p = 0.029$). In same-gender rounds, all-female groups succeed more often than all-male groups among *MixedFirst* participants, by 10.3 percentage points ($p < 0.001$), but not among *SameFirst* participants.

Because successful coordination requires only one member to volunteer, higher success rates in female-heavier groups without higher average individual volunteering are consistent with more effective stabilization of group outcomes rather than a generally stronger preference for volunteering. The advantage of all-female groups among *MixedFirst* participants also mirrors the within-person increase among *MixedFirst* women reported in Table 1.

[14] *MixedFirst* men also show some movement toward the near-always state, from 19 to 31 percent. A transition test distinguishes this pattern from that of *MixedFirst* women. Entries into the near-always state relative to exits are 11:0 among *MixedFirst* women (exact McNemar $p = 0.001$) and 6:2 among *MixedFirst* men ($p = 0.29$). Among men, the share in the never-volunteer state also rises, with five entries and one exit ($p = 0.22$). Their pattern is therefore better described as statistically insignificant two-sided movement, in contrast to the one-directional change among women. All *SameFirst* transition counts are balanced.

Table 5: Group-level coordination success, by group gender composition

	Mixed groups (1)	Same-gender groups (2)
Panel A. Average adjusted success probabilities		
SameFirst × 2F1M / all-female	0.797	0.752
SameFirst × 1F2M / all-male	0.643	0.801
MixedFirst × 2F1M / all-female	0.834	0.814
MixedFirst × 1F2M / all-male	0.720	0.711
Panel B. Contrasts		
Composition effect, SameFirst	$+0.1541^{***}$	-0.0488
	(0.001)	(0.651)
Composition effect, MixedFirst	$+0.1143^{**}$	$+0.1028^{***}$
	(0.029)	(0.000)
SameFirst effect, 2F1M / all-female groups	-0.0370	-0.0622
	(0.281)	(0.518)
SameFirst effect, 1F2M / all-male groups	-0.0768	$+0.0894$
	(0.231)	(0.224)
Difference-in-differences	$+0.0399$	-0.1516
	(0.565)	(0.170)
Group-round observations	400	300

Notes: Success equals one if at least one group member volunteers. Row labels of the form "*x* / *y*" refer to columns (1) and (2), respectively. In column (1), mixed groups consist of two women and one man (2F1M) or one woman and two men (1F2M); in column (2), same-gender groups are all-female or all-male. The composition effect is the difference in success probability between the female-heavier and the other composition: 2F1M versus 1F2M in column (1), and all-female versus all-male in column (2). The SameFirst-effect rows compare *SameFirst* and *MixedFirst* while holding composition fixed. Estimates are from probit models including round controls. Parentheses report p-values based on delta-method standard errors clustered at the session level. Because there are nine session clusters, the p-values should be interpreted cautiously. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

These group-level differences are among the statistically strongest patterns in the data, but they should be interpreted cautiously. Treatment and composition vary at the session and group levels, and the effective number of session clusters is only nine. The findings are consistent with women stabilizing group coordination but do not distinguish whether the relevant explanation is belief-related or preference-based.

5.4 Institutional Implications

The findings offer a complementary perspective on gender differences in the allocation of LPTs. Prior research emphasizes external social expectations that women should volunteer. In our sample, however, the hypothetical assignment task does not reproduce a female-directed assignment tendency. Instead, the observed behavior is compatible with two belief-related considerations—strategic uncertainty and revised empirical expectations—and one preference-based consideration, gender in-group favoritism.

Although these considerations differ conceptually, they share an institutional implication. Under an opt-in allocation system, the task is more likely to fall to members who are especially concerned about coordination failure, who place greater weight on other group members' payoffs, or who do not expect someone else to act. Conversely, members who believe that someone else will volunteer have an incentive to free-ride. Gender composition may influence each of these forces, allowing voluntary allocation to generate unequal task burdens even when participants do not explicitly assign the task to women.

Identifying which consideration dominates is therefore not necessary for the basic institutional implication. All three become behaviorally consequential because the task is allocated through an unresolved coordination problem. Simply encouraging participants who volunteer frequently to decline the task may not solve that problem. If beliefs and allocation procedures remain unchanged, unilateral refusal can result in the task being left undone rather than being reassigned more efficiently.

This distinction matters for organizational policy. If unequal LPT allocation were driven exclusively by an explicit female-directed norm, interventions might focus primarily on individual behavior or attempts to change that norm, such as encouraging women to decline requests (Sandberg, 2013). If volunteering also reflects responses to strategic uncertainty, expectations about others' actions, or concern for group members' payoffs, interventions focused only on individual refusal are incomplete. Structural allocation rules are required to ensure that task completion does not depend on which employee is most willing to resolve the coordination problem.

One direct intervention is to replace opt-in volunteering with a transparent assignment mechanism. Rotational assignment, explicit managerial designation, or another predetermined allocation rule can remove uncertainty about whether the task will be completed and reduce the opportunity for belief-based free-riding. Such rules can also prevent the burden from varying mechanically with team gender composition. More generally, this argument is consistent with evidence that choice architecture can shape gender disparities in competitive and organizational decisions (He et al., 2021).

Formal assignment may also improve allocative efficiency. Voluntary coordination can direct low-promotability work toward employees who are most concerned about group failure rather than toward those for whom completing the task has the lowest opportunity cost. If highly productive employees repeatedly absorb low-value coordination tasks, the organization may misallocate human capital (Bircan et al., 2025). Transparent assignment systems are therefore potentially valuable not only as equity interventions but also as mechanisms for preventing productive employees from being systematically diverted toward low-promotability work.

5.5 Limitations and Future Research

Several limitations qualify these interpretations. First, the laboratory setting abstracts from explicit task requests, repeated interpersonal relationships, managerial hierarchies, and reputational consequences that characterize workplace LPT allocation. Participants were university students recruited in South Korea, so the setting is particularly useful for examining whether previously documented patterns extend across contexts, but the findings should not be generalized mechanically to other populations or institutions.

Second, none of the three candidate considerations is directly identified. The experiment does not elicit phase-specific beliefs about whether others will volunteer, manipulate the precision of those beliefs, or measure gender identification and payoff weights. The post-experiment expectation measures are collected after the volunteering decisions and may therefore reflect retrospective rationalization. Evidence concerning gender-dependent social preferences is also mixed (Ben-Ner et al., 2004; Sutter et al., 2009). The proposed interpretations should consequently be viewed as organizing frameworks rather than established mechanisms.

Third, the sample of 150 participants limits statistical precision, and several estimates are directionally informative but imprecise; minimum detectable effects for the main contrasts are on the order of 20 percentage points. The group-level analysis also relies on only nine session clusters. In addition, the treatment comparisons combine prior exposure with the order in which environments are experienced: mixed-gender rounds occur later for *SameFirst* participants than for *MixedFirst* participants. Difference-in-differences comparisons absorb gender-invariant order effects but cannot rule out gender-specific order effects.

Finally, the assignment task is hypothetical and unincentivized. It provides evidence about stated assignment choices but does not directly measure incentivized normative expectations, empirical expectations about who will volunteer, or actual supervisory behavior.

Future research could distinguish the three considerations more directly by eliciting incentivized first- and second-order beliefs during each phase, independently manipulating the precision and level of those beliefs, and measuring gender identification or allocation choices toward same- and opposite-gender recipients. Designs that counterbalance transition order or hold group composition fixed across otherwise comparable treatments would help separate the effects of prior exposure, composition, and accumulated experience. Field studies could then examine whether similar responses occur when LPTs are explicitly requested, reputational consequences are present, and task allocation takes place within continuing organizational relationships.

6 Conclusions

This paper investigates whether experiences in same-gender environments shape later willingness to volunteer for low-promotability tasks (LPTs) in mixed-gender groups. Using a sequential laboratory experiment, we find no average gender gap in volunteering in mixed-gender groups, and an asymmetric response to prior same-gender exposure: men reduce volunteering after transitioning into mixed-gender groups, while women maintain theirs. The resulting gender gap after same-gender exposure is economically sizable but not statistically significant at conventional levels.

These patterns do not reflect gendered social expectations about task assignment. In hypothetical assignment tasks, participants of both genders predominantly assign LPTs to men, contrary to the expectation-based mechanism that dominates the existing literature. We discuss differential aversion to strategic uncertainty as one candidate mechanism consistent with several of these patterns, while noting that accounts based on revised expectations about others' behavior and in-group favoritism fit parts of the data as well; separating these channels requires designs with direct belief elicitation.

By integrating insights from the literature on LPTs, same-gender environments, strategic uncertainty, and gender-dependent social preferences, this study broadens the theoretical understanding of gender differences in organizational behavior. It highlights that early social environments shape not only immediate behavior but also how individuals respond to uncertainty in later, more heterogeneous settings. From a policy perspective, the findings suggest that reducing ambiguity about responsibility and coordination, rather than focusing solely on social norms, may be an effective approach to mitigating unequal task burdens in organizations.

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A Further Results

Table A.1: Balance Test: Treatment Randomization Check

Variable	SameFirst	MixedFirst	Diff	P-value
Female	0.50 (0.50)	0.50 (0.50)	0.00	1.000
Age	22.41 (3.08)	22.24 (2.73)	0.17	0.714
experience	1.92 (0.27)	1.94 (0.23)	-0.02	0.601
nFriends	1.19 (0.40)	1.04 (0.20)	0.15	0.004*
Girl's School (women only)	0.41 (0.50)	0.42 (0.50)	-0.01	0.956
Boy's School (men only)	0.62 (0.49)	0.53 (0.50)	0.09	0.451
selfPressure	2.13 (1.35)	2.21 (1.21)	-0.08	0.702
sameGenExp	2.74 (1.41)	2.83 (1.42)	-0.09	0.699
concernOthers	1.87 (1.22)	1.90 (1.29)	-0.03	0.880
volunteer	3.49 (1.15)	3.32 (1.25)	0.17	0.396
leader	2.78 (1.40)	2.58 (1.30)	0.20	0.369
socialConcern	3.42 (1.12)	3.47 (1.17)	-0.05	0.794
nonConfirmity	2.99 (1.21)	2.86 (1.40)	0.13	0.557

Notes: Experience is coded 1 = has participated in an economics experiment before, 2 = has not; nFriends is an ordinal category (1 = none, 2 = one or two, 3 = three or more acquaintances in the session). Welch *t*-tests.

Table A.2: Rounds 7 and 8 only: cell probabilities and contrasts

	(1)	(2)
Panel A. Average adjusted probabilities		
SameFirst × Female	0.359	0.393
SameFirst × Male	0.320	0.293
MixedFirst × Female	0.417	0.393
MixedFirst × Male	0.444	0.466
Panel B. Contrasts		
SameFirst effect, women	−0.0576	−0.0002
	(0.514)	(0.998)
SameFirst effect, men	−0.1241	−0.1725**
	(0.196)	(0.028)
Gender gap, SameFirst	+0.0386	+0.0994
	(0.663)	(0.224)
Gender gap, MixedFirst	−0.0278	−0.0729
	(0.771)	(0.400)
Difference-in-differences	+0.0664	+0.1723
	(0.611)	(0.139)
Survey controls	No	Yes
Observations	300	300

Notes: Both treatments are observed at identical rounds (7–8): *SameFirst* participants are in their first two mixed-gender rounds, *MixedFirst* participants in their last two. Computation of cell probabilities, contrasts, and p-values as in Table 2. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Marginal Effects of Investment: *SameFirst* Treatment

	Full (1–14R)		Phase 1 (Same, 1–6R)		Phase 2 (Mixed, 7–14R)	
	(1)	(2)	(3)	(4)	(5)	(6)
Female	0.0604	0.0874	0.0299	0.0635	0.0832	0.1045
	(0.4172)	(0.1950)	(0.6940)	(0.3610)	(0.2901)	(0.1480)
Round	0.0003	0.0004	0.0026	0.0023	0.0072	0.0072
	(0.9269)	(0.9060)	(0.8282)	(0.8430)	(0.2173)	(0.2090)
Controls	No	Yes	No	Yes	No	Yes
selfPressure	-	0.0722***	-	0.0940***	-	0.0556*
concernOthers	-	0.0620*	-	0.0414	-	0.0765**
Observations	1092	1092	468	468	624	624

Notes: Dependent variable: individual investment decision (1–invest, 0–don't invest). Survey controls include selfPressure, sameGenExp, concernOthers, volunteer, leader, socialConcern, nonConfirmity. The table presents marginal effects of Probit regression. Standard errors are clustered at the individual level. p-values are reported in parentheses.

Table A.4: Marginal Effects of Investment: MixedFirst Treatment

	Full (1–14R)		Phase 1 (Mixed, 1–8R)		Phase 2 (Same, 9–14R)	
	(1)	(2)	(3)	(4)	(5)	(6)
Female	0.0434	0.0028	-0.0105	-0.0418	0.1158	0.0638
	(0.5534)	(0.9640)	(0.8891)	(0.5310)	(0.2237)	(0.4350)
Round	0.0060	0.0061	0.0038	0.0037	-0.0115	-0.0110
	(0.2106)	(0.2060)	(0.6287)	(0.6350)	(0.2611)	(0.2770)
Controls	No	Yes	No	Yes	No	Yes
selfPressure	-	0.1161^{***}	-	0.0957^{***}	-	0.1438^{***}
concernOthers	-	-0.0392^{*}	-	-0.0193	-	-0.0661^{**}
Observations	1008	1008	576	576	432	432

Notes: Dependent variable: individual investment decision (1–invest, 0–don't invest). Survey controls include selfPressure, sameGenExp, concernOthers, volunteer, leader, socialConcern, nonConfirmity. The table presents marginal effects. Standard errors are clustered at the individual level. p-values are reported in parentheses.

Table A.5: Investment of male participants, by single-sex schooling background

	Pooled Data (All Rounds)		Same Environments Only		Mixed Environments Only	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Average adjusted probabilities						
SameFirst × Boy School	0.393	0.349	0.414	0.375	0.367	0.326
SameFirst × Co-ed	0.248	0.255	0.231	0.243	0.252	0.260
MixedFirst × Boy School	0.459	0.481	0.492	0.522	0.444	0.456
MixedFirst × Co-ed	0.307	0.340	0.301	0.323	0.321	0.361
Panel B. Contrasts						
SameFirst effect, Boy-School men	−0.0659	−0.1318	−0.0788	−0.1470	−0.0767	−0.1308
	(0.548)	(0.194)	(0.607)	(0.324)	(0.539)	(0.241)
SameFirst effect, Co-ed men	−0.0589	−0.0844	−0.0691	−0.0799	−0.0682	−0.1009
	(0.573)	(0.292)	(0.640)	(0.517)	(0.541)	(0.298)
Boy School effect, SameFirst	+0.1451	+0.0936	$+0.1823^{*}$	+0.1326	+0.1150	+0.0652
	(0.152)	(0.288)	(0.093)	(0.148)	(0.270)	(0.503)
Boy School effect, MixedFirst	+0.1520	+0.1409	+0.1919	$+0.1998^{*}$	+0.1235	+0.0950
	(0.178)	(0.146)	(0.152)	(0.060)	(0.304)	(0.401)
Difference-in-differences	−0.0069	−0.0473	−0.0096	−0.0671	−0.0085	−0.0298
	(0.964)	(0.732)	(0.955)	(0.655)	(0.957)	(0.841)
Survey controls	No	Yes	No	Yes	No	Yes
Observations	1,050	1,050	450	450	600	600

Notes: Male participants only. Boy School indicates graduation from an all-boys middle or high school. Specifications and computation of cell probabilities, contrasts, and p-values as in Table 3. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Volunteer types, prospective splits, and stated expectations

	Phase 1		Phase 2	
	Near-always	Never	Near-always	Never
Panel A. Number (percent) of volunteer types by cell				
SameFirst men ($n = 39$)	7 (18)	12 (31)	6 (15)	12 (31)
SameFirst women ($n = 39$)	6 (15)	10 (26)	6 (15)	11 (28)
MixedFirst men ($n = 36$)	7 (19)	12 (33)	11 (31)	16 (44)
MixedFirst women ($n = 36$)	2 (6)	5 (14)	**13 (36)**	7 (19)
	Phase 1 (%)	Phase 2 (%)	Δ (pp)	p
Panel B. Prospective split by mixed-phase behavior, women only				
MixedFirst, above median ($n = 14$)	63.4	71.4	+8.0	0.358
MixedFirst, below median ($n = 22$)	21.6	36.4	+14.8	0.093
SameFirst, above median ($n = 18$)	66.7	68.1	+1.4	0.708
SameFirst, below median ($n = 21$)	15.1	16.7	+1.6	0.584
	n	Phase 1 (%)	Phase 2 (%)	sameGenExp (1–5)
Panel C. Phase-2 volunteer types among MixedFirst women				
Near-always ($\geq 5/6$)	13	51.0	93.6	2.38
Intermediate	16	33.6	36.5	3.00
Never	7	23.2	0.0	3.43

Notes: Panel A reports the number of participants of each type, with cell percentages in parentheses; Near-always denotes an individual investment rate of at least 5/6 within the phase (at least five of six rounds in same-gender rounds, or seven of eight in mixed-gender rounds), Never a rate of zero. In Panel B, women are classified by their own-treatment median investment rate in Phase 1, before Phase-2 behavior is observed; entries are mean investment rates, Δ is the average within-person change, and p-values are from paired t-tests. The *SameFirst* rows serve as a mean-reversion placebo. Panel C reports mean investment rates by phase and mean agreement that someone of one's own gender would invest (elicited after the decision rounds), by phase-2 type; the gradient is monotone (Spearman $p = 0.11$), absent among *SameFirst* women, and may partly rationalize rather than cause behavior. Transitions into the near-always state (entries:exits) are 11:0 for *MixedFirst* women (exact McNemar $p = 0.001$), 6:2 for *MixedFirst* men ($p = 0.29$), and balanced in both *SameFirst* cells. Phase lengths differ across treatments; restricting both phases to their first six rounds leaves the *MixedFirst*-women results unchanged (11:0, $p = 0.001$) and further attenuates the movement among *MixedFirst* men (6:4, $p = 0.75$). All statistics are exploratory.

B Appendix: Experimental Instructions

(*Instructions for the *SameFirst* are translated in English. Original (Korean) instructions are available upon request.)

Welcome

Thank you for participating in this experiment. Before we get started, please answer the following questions.

[*Participants answer their gender, birth year, major, previous experience in economics experiment participation, and a number of recognizable friends who happen to join the same session.*]

Please read carefully the following experiment instructions. Comprehension check questions to make sure you understood the instructions will be followed.

The payoffs from this experiment will depend on your choices, other participants' choices, and luck. The currency unit used for this experiment is a "token."

Experiment Overview

The main part of the experiment consists of two sessions, and you will choose your decision for 14 rounds. Each round's decisions are anonymous, and you make a decision within a randomly-formed group of three. Some conditions may vary in each decision round, so please pay attention.

Main Task: Decide to Invest or Not

Participants are randomly assigned to a new group of three people at the beginning of each round. After that, each group member simultaneously decides to **invest** or **not**.

- If none of the three people invests, each person receives 100 tokens.
- If one person decides to invest and the other two do not, the investor receives 125 tokens, and the two who did not invest each receive 200 tokens.
- If two or more people decide to invest, only one person is randomly selected from among them to actually invest. In this case as well, the person who invests receives 125 tokens, and the two who do not invest receive 200 tokens.

Example Suppose that among A, B, and C, A and B decide to invest, and C decides not to invest. In this case,

- One person is randomly selected from A and B to be the investor.
- If A becomes the investor: A receives 125 tokens, and B and C each receive 200 tokens.
- If B becomes the investor: B receives 125 tokens, and A and C each receive 200 tokens.

Feedback at the End of Each Round At the end of each round, participants are only informed that the round has ended. Information about the group members' choices and the payoff outcomes will be provided only after the experiment ends, and only for one randomly selected round.

Payment After the experiment ends, the computer randomly selects one round. The tokens earned in the selected round are converted into cash at a rate of 100 KRW per token and paid out. Since every round has an equal chance of being selected, it is advantageous to participate carefully in every round.

Comprehension Quiz You must answer all quiz questions correctly in order to proceed to the next stage. If necessary, please review the explanation above again.

Q1 Which of the following statements is incorrect?

(1) If all group members do not invest, each receives 100 tokens.
(2) If I decide to invest, I always receive 125 tokens.
(3) If I do not invest but someone else decides to invest, I receive 200 tokens.
(4) Group members are newly formed in every round.

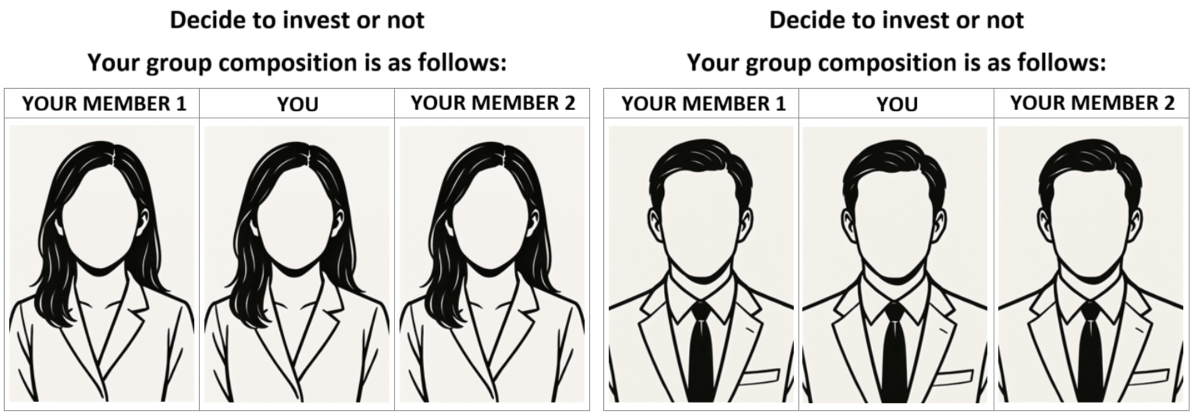
Q2 Suppose that all group members, including yourself, decide to invest. Which of the following statements is correct?

(1) Everyone receives 100 tokens.
(2) Everyone receives 125 tokens.
(3) Two members receive 125 tokens, and the remaining one receives 200 tokens.
(4) One member receives 125 tokens, and the remaining two receive 200 tokens.

Section 1

In this section, you will repeat the investment decision described above six times. In each round, a new group of three people is randomly formed, consisting only of participants of the same gender as you.

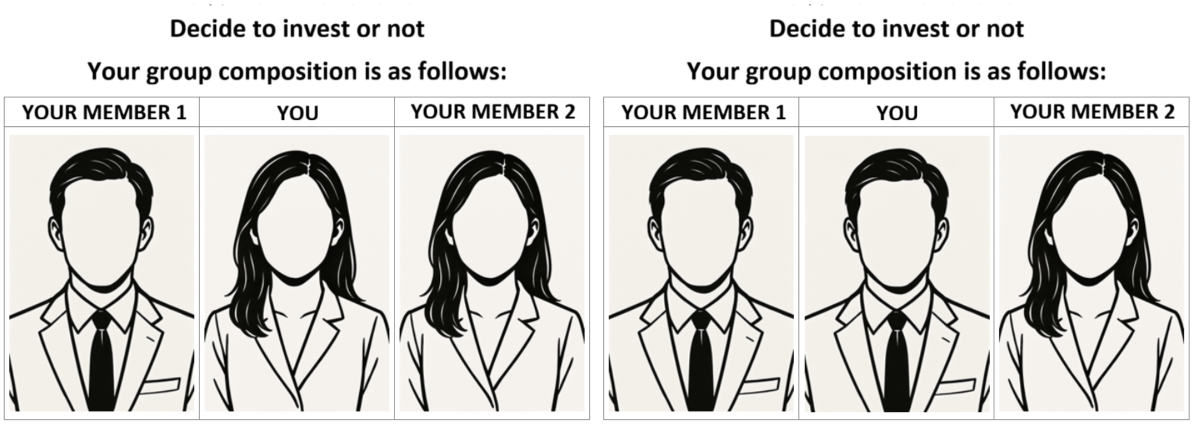
[*In each round of Section 1, participants see three silhouette images of the same gender of themselves, as follows.*]



Section 2

In this section, you will repeat the investment decision described above eight times. In each round, a new group of three people is randomly formed, and the group will include at least one participant of a different gender from you.

[*In each round of Section 2, participants see three silhouette images of at least one different gender of themselves. Two examples are as follows.*]



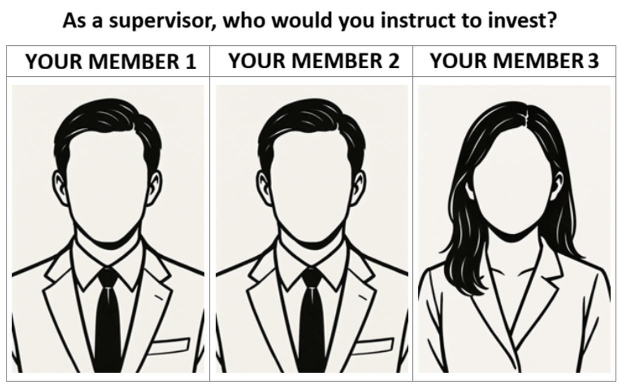
[*At the end of round 14, each participant's payoff of one randomly selected round is informed.*]

Post-experiment survey

Review of the experiment Recall the experiment you had just participated, and answer how much you agree with the following statements. [*Likert scale from 1 (completely disagree) to 5 (completely agree) is followed.*]

- "I feel pressure to invest."
- "I expect someone else with the same gender of mine would invest."
- "I care about how others regard my decisions."

If you were in a position as a supervisor of the group and had to instruct one of the group members to make an investment, which of the following would you choose?



If you were in a position as a supervisor of the group and had to instruct one of the group members to make an investment, which of the following would you choose?

Individual characteristics Answer how much you agree with the following statements. [*Likert scale from 1 (completely disagree) to 5 (completely agree) is followed.*]

- "If a task has to be done, I tend to volunteer even when I'm reluctant."

- "I enjoy playing a leader role."
- "I respond to the social expectations."
- "I argue with friends who have different opinions."

Choose the types of the middle and high schools you graduated. [*Choice options are co-educational school, single-gender school, other/hard to describe.*]